

SkillClaw

Let Skills Evolve Collectively with Agentic Evolver

Collective Skill Evolution for Multi-User LLM Agent Ecosystems

arxiv.org/abs/2604.08377



Agent Skills Remain Static After Deployment

Users Rediscover Solutions Independently

STATIC PARADIGM

- Skills are manually installed from a centralized hub
- Solutions discovered during interaction do not persist beyond individual sessions
- Same patterns of failure and recovery are repeatedly observed across users
- Each user is forced to rediscover solutions independently
- No system-level knowledge accumulation

PRIOR APPROACHES FALL SHORT

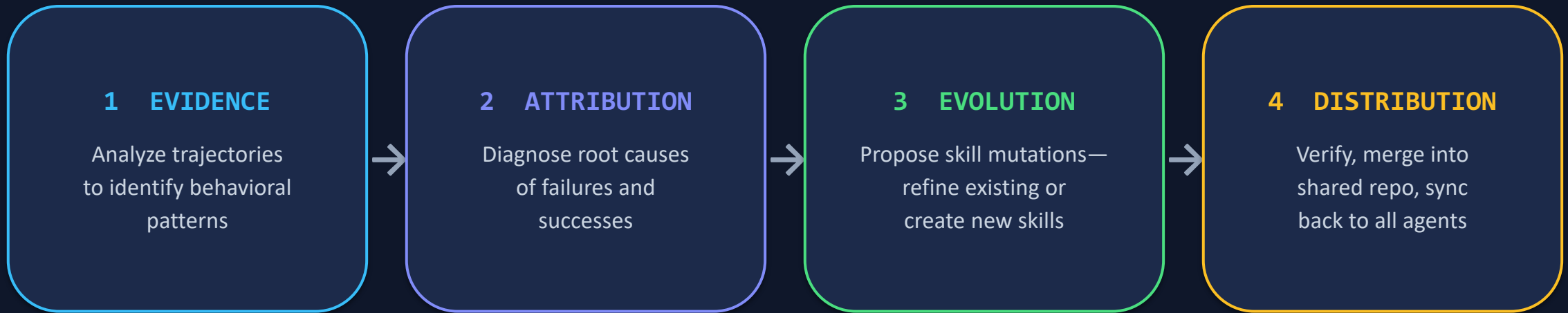
Memory-based methods

Store past trajectories but remain tied to specific instances; hard to generalize

Skill-based methods

Compress experience into structured instructions but treat the library as static — it never evolves through usage

Closed-Loop Pipeline: Multi-User Interaction → Shared Skill Evolution



↻ Continuous feedback loop – evolved skills generate new trajectories in next cycle

MULTI-USER INPUT

Independent agents produce structured session trajectories preserving full action–feedback causal chains. Trajectories are aggregated across users and grouped by referenced skills.

AGENTIC REASONING

Evolution is driven by an autonomous agent performing open-ended reasoning over interaction evidence — not predefined update rules.

Agentic Evolver: Evidence → Attribution → Evolution

Three-Stage Pipeline Driven by Open-Ended Agent Reasoning

3

UTION

Propose Skill Mutations

Refine existing skills or create entirely new ones based on attributed evidence. Generate candidates that are verified before merging into the shared repository.

⚡ **Key distinction:** Skill updates are produced through open-ended reasoning, not predefined update rules or fixed heuristics.

Cross-User Trajectory Aggregation

Exposing Skill Behavioral Boundaries Through Diverse Contexts

Different users exercising the same skill under diverse contexts produce complementary views of that skill's behavioral boundary — revealing both conditions under which it works and those under which it breaks.
A single user rarely generates enough signal to separate a generalizable improvement from an idiosyncratic fix.

COMPATIBLE CLAW-STYLE SYSTEMS

OpenClaw

CoPaw

IronClaw

PicoClaw

ZeroClaw

NanoClaw

NemoClaw

SUCCESS PATTERNS

Trajectories where the skill works correctly under various conditions provide evidence for generalizable best practices.

FAILURE MODES

Trajectories where the skill fails expose edge cases and boundary conditions that no single user would encounter alone.

Case Study: Multimodal Creative Pipeline Skill

6 Days of Evolution Attempts — Conservative Verification in Action

Day	Candidate Skill	Function / Change	Outcome	Next-Day Action
1	validate-tmp-workspace-inputs	Check /tmp_workspace inputs & env setup	Accept	Upgrade to new best pool
2	multimodal-input-validation-and-setup	Multimodal input validation & output env init	Reject	Keep current best pool
3	multimodal-creative-task-pipeline	Creative pipeline (extract from PDF/video/image)	Reject	Keep current best pool
4	multimodal-creative-task-pipeline (impr.)	Added image classification, visual generation	Reject	Keep current best pool
5	multimodal-creative-task-pipeline + validate	Creative pipeline + per-file fail-fast validation	Reject	Keep current best pool
6	multimodal-creative-task-pipeline (cand.)	Extended PDF-to-poster / document-to-visual	Reject	Not admitted to next cycle

RESULT: 1 OF 6 ACCEPTED

The conservative verification process prevents regressions. Only candidates that demonstrably improve the skill pool pass validation — the system favors stability over aggressive mutation.

Case Study: Git Push Auth-Fallback Skill

Iterative Refinement Over 6 Days — Successful Evolution Before Plateau

Day	Candidate Skill	Change Summary	Outcome	Next-Day Action
1	git-push-with-auth-fallback	Safe fallback instead of blocking on push failure	Accept	Add to Safety best pool
2	git-push-with-auth-fallback	Unified patch/bundle filenames, reduced inconsistency	Accept	Keep updated best pool
3	git-push-with-auth-fallback + git-clone	Auth-alternative paths & secrets audit; fixed subshell	Accept	Keep current best pool
4	(none — retest)	Same-pool retest; validator confirmed current as best	Accept	Continue same best pool
5	git-push-with-auth-fallback	Added "push hang = auth failure"; no improvement	Reject	Keep current best pool
6	git-push-with-auth-fallback	Added identity config & filename consistency	Reject	Not admitted to next cycle

RESULT: 3 OF 6 ACCEPTED

When evidence supports iterative refinement, the system allows progressive improvement. The skill was successfully improved 3 times before reaching a natural plateau — further mutations added complexity without measurable benefit.

Three Properties That Distinguish SkillClaw

AGENTIC EVOLUTION

Skill updates are produced through open-ended reasoning over evidence rather than predefined update rules. The evolver is itself an agent.

IMPLICATIONS

- Agent systems can transition from static skill libraries to continuously improving ecosystems through collective interaction data
- Evaluated on WildClawBench with Qwen3-Max backbone — demonstrates substantial improvements across tasks